

LAPORAN PRATIKUM
PRAKTIKUM ELEKTRONIKA DIGITAL

Dosen Pengampu:

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Disusun Oleh:

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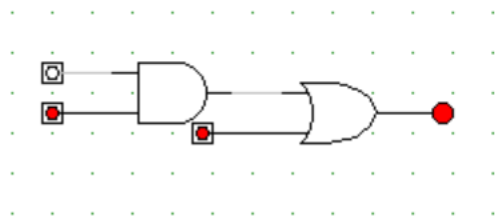
PROGRAM STUDI
TEKNIK INFORMATIKA
FAKULTAS SAINS DAN TEKNOLOGI
UNIVERSITAS ISLAM NEGERI MAULANA MAILK IBRAHIM MALANG
2024

$$X = \bar{A}B + \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + A\bar{B}C$$

	00	01	11	10
0	1	1	1	1
1		1		

Truth table: $\bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C$

Penyederhanaan aljabar Boolean : $C' + A'B$



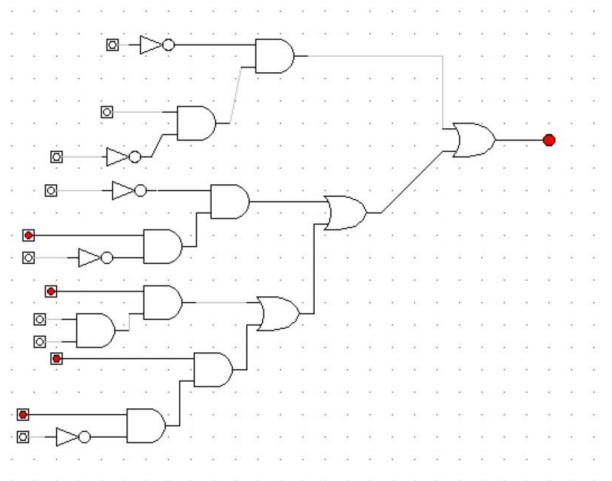
$$X = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}B\bar{C}D + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + ABCD$$

	00	01	11	10
00		1		
01		1		
11	1	1	1	
10		1		

TRUTH TABLE: $\bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}B\bar{C}D + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + ABCD$

$$\bar{C}D\bar{B} + \bar{C}D\bar{A} + ABD + A\bar{B}\bar{C}$$

PENYEDERHANAAN ALJABAR BOOLEAN:



$$X = \overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}D + A\overline{B}\overline{C}D + \overline{A}BCD + ABCD$$

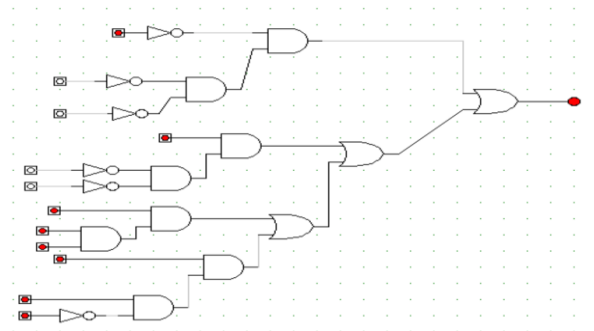
	00	01	11	10
00				
01	1	1	1	
11	1	1	1	
10				

TRUTH TABLE :

$$\overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}D + \overline{A}BCD + A\overline{B}\overline{C}\overline{D} + A\overline{B}\overline{C}D + ABCD$$

PENYEDERHANAAN ALJABAR BOOLEAN :

$$B\overline{C}\overline{A} + B\overline{C}\overline{D} + BDC + BDA$$



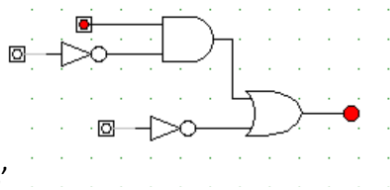
$$X = \overline{A}\overline{B}\overline{C} + \overline{A}\overline{C}\overline{D} + \overline{A}\overline{B} + A\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}$$

	00	01	11	10
00	1	1	1	1
01				
11	1			1
10	1	1	1	1

TRUTH TABLE

$$\overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}D$$

PENYEDERHANAAN ALJABAR BOOLEAN :



$AD' + B'$

$$X = \overline{A}\overline{D} + \overline{A}\overline{B}\overline{D} + \overline{A}\overline{C}D + \overline{A}CD$$

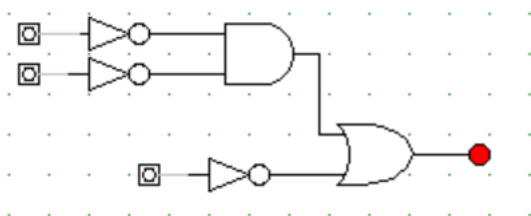
	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11				
10	1			1

TRUTH TABLE:

$$\overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}C\overline{D} + \overline{A}\overline{B}C\overline{D} + \overline{A}\overline{B}C\overline{D} + \overline{A}\overline{B}C\overline{D} + \overline{A}\overline{B}C\overline{D} + \overline{A}\overline{B}C\overline{D} + \overline{A}\overline{B}C\overline{D} + \overline{A}\overline{B}C\overline{D} + \overline{A}\overline{B}C\overline{D}$$

PENYEDERHANAAN ALJABAR BOOLEAN

$$\overline{D}\overline{B} + \overline{A}$$



$$X = \overline{A}\overline{B}\overline{D} + \overline{A}\overline{C}\overline{D} + \overline{A}B\overline{C} + \overline{A}B\overline{C}D + \overline{A}B\overline{C}\overline{D}$$

TRUTH TABLE:

$$\overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}C\overline{D} + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D + A\overline{B}\overline{C}\overline{D} + A\overline{B}\overline{C}D$$

$$B\overline{C} + A\overline{C}\overline{D} + \overline{B}DC + \overline{B}D\overline{A}$$

PENYEDERHANAAN ALJABAR BOOLEAN:

